

Methodology Comparison: AIOLOS, Taylor et al. (2025, 2026), and Xing et al. (2023), and Their Adoption Potential for EXHALE

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Status (dated note added 2026-08-15). Written before 2026-07; the EXHALE column and the adoption assessment predate the metal, molecular and diffusion work. Since implemented: the temperature-dependent Taylor et al. (2025) Penning rate (unconditional default), He/H (+ metal) diffusive separation (`binary_element_diffusion.f90`, keys `He_diffusion` / `He_metal_diffusion`, default off), molecular chemistry $\text{H}_2/\text{H}_2^+/\text{H}_3^+/\text{HeH}^+$ with H_3^+ infrared cooling and the lower-atmosphere tiers (`lower_atmosphere/`), charge exchange re-sourced to Huang et al. (2023) Table 4, and 10 elements / 27 metal ion stages. The text below is kept as the record of what the code looked like at the time; the markdown edition carries the same note with inline markers.

Abstract

This note compares the methodology of three independent modeling efforts for XUV-driven atmospheric escape from close-in exoplanets, and assesses which elements could be adopted into EXHALE (our extended ATEs fork). The efforts are the general-purpose 1-D radiation-hydrodynamics code AIOLOS (Schulik & Booth 2023); the dedicated thermosphere–ionosphere escape model of Taylor et al. (2025, 2026), built on the Koskinen et al. code; and the multi-fluid PLUTO-based model of Xing et al. (2023). EXHALE’s own methodology is summarized so the adoption analysis has a concrete baseline. The central finding is that EXHALE’s largest physical gap is the absence of any radius-dependent He/H separation, and that a Taylor-style single-fluid diffusion term (plus a temperature-dependent Penning ionization rate) is the highest-value, lowest-risk upgrade. [2026-08-15: both have since been implemented — see the status note above.]

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1 Scope and the codes considered

We compare three efforts for modeling XUV-driven escape from close-in planets:

- **Aiolos** — the general-purpose 1-D multi-species radiation-hydrodynamics code of Schulik & Booth (2023, “SB23”), distributed on GitHub (github.com/Schulik/aiolos).
- **Taylor et al. (2025)**, *ApJ* 989:68 — “A Multispecies Atmospheric Escape Model with Excited Hydrogen and Helium: Application to HD209458b,” built on the Koskinen et al. (2013a,b; 2022) thermosphere–ionosphere code.
- **Taylor et al. (2026)**, *ApJ* 999:214 — “Helium Escape in Context: Comparative Signatures of Four Close-in Exoplanets,” a direct application/extension of Taylor (2025) to HD 209458b, HD 189733b, HD 149026b, and GJ 1214b.
- **Xing et al. (2023)**, *ApJ* 953:166 — “The Mass Fractionation of Helium in the Escaping Atmosphere of HD 209458b,” a *multi-fluid* model built inside PLUTO.

“Taylor” denotes the shared Koskinen-based framework used in both Taylor papers unless a specific paper is named.

2 One-paragraph characterization of each code

Aiolos. A *general-purpose* radiation-hydrodynamics code, not a dedicated escape code. It solves the time-dependent Euler equations (mass, momentum, energy) for an arbitrary number of species — each a **separate fluid** with its own density, momentum, and internal energy — coupled by an inter-species **friction/drag** solver. It uses finite-volume **HLLC** fluxes with **PLM** (MC-limited) reconstruction and second-order TVD Runge–Kutta time-stepping, on Cartesian/cylindrical/**spherical** log-spaced grids. Radiation is treated with **flux-limited diffusion (FLD)** over arbitrary in/out spectral bands, solved implicitly via a block-tridiagonal system; chemistry ranges from a fast C2Ray ionization scheme to a full implicit multi-reaction network. Escape is one of many problems it can run.

Taylor (2025/2026). A *dedicated* 1-D spherical thermosphere–ionosphere + hydrodynamic-escape model (Koskinen et al. 2013/2022), integrated to steady state, with a **single bulk momentum equation** but **multi-species molecular + eddy diffusion** that redistributes species (so He and H can separate diffusively). Its distinguishing strengths are physical completeness of the *microphysics*: a full He(2^3S) and H($n=2$) non-LTE network solved by the KPP kinetic pre-processor, a **Ly α Monte Carlo** radiative-transfer model (Huang et al. 2017) iterated with the hydro solution for H α , a new high-resolution He(2^3S) photoionization cross section (B-spline K-matrix), temperature-dependent Penning ionization, and self-consistent coupling to a photochemical lower/middle atmosphere at the μbar level.

Xing (2023). A *dedicated* 1-D spherical escape model whose distinguishing feature is that it is genuinely **multi-fluid**: H, H⁺, He, He⁺, and e[−] each carry their **own velocity**, coupled by resonant/non-resonant collisional drag. This is the only one of the three that can capture **He/H mass fractionation** (heavy He under-dragged by escaping H) self-consistently in the dynamics rather than through a diffusion coefficient. It is built on PLUTO (HLL Riemann solver, 3rd-order TVD RK) with a novel Riemann split to carry the electron/electric-field pressure term. The He(2³S) 10830 Å line is post-processed, not solved in the hydro loop.

Exhale (baseline for comparison). A 1-D spherical **single-fluid** steady-state escape code (ATES heritage), with a two-stage PLM→WENO3 reconstruction and an approximate Riemann solver, integrated to steady state on a mass-flux ($\rho v r^2$) convergence criterion. Ionization of H, He, and trace metals (C/N/O) is solved as one coupled nonlinear system (analytic-Jacobian Newton with a MINPACK `hybrd1` fallback). It already includes a He(2³S) metastable network, an H($n=2$)/Balmer non-LTE population (Christie et al. 2013), a Ly α radiative-transfer suppression via a Neufeld escape-probability treatment, Badnell RR+DR recombination, Voronov collisional ionization, Kingdon & Ferland charge exchange, CHIANTI metal-line cooling (Mg/Ca/Na/Fe II, C/N/O), a Roche/tidal potential (RLOF), and a transmission-spectrum post-processor (`EXHALE_transit.py`).

3 Side-by-side methodology table

Axis	Aiolos (SB23)	Taylor 25/26	Xing 23	Exhale (current)
Purpose	General multi-species RHD	Dedicated escape + line diagnostics	Dedicated escape, He/H fractionation	Dedicated escape + line diagnostics
Dimensionality	1-D (sph/cyl/cart)	1-D spherical	1-D spherical	1-D spherical
Time	Time-dep. → steady	Time-dep. → steady	Time-dep. → steady	Steady-state (relaxation)
Fluid model	Multi-fluid + friction	Single bulk + diffusion	Multi-fluid	Single fluid
He/H separation	Via friction/drag	Via molecular+eddy diffusion	Via multi-fluid dynamics	None (fixed He/H [2026-08-15: since implemented])
Reconstruction	PLM + MC limiter	finite-difference	PLM (PLUTO)	PLM→WENO3
Riemann solver	HLLC	—	HLL (+ e [−] pressure split)	Approximate (ATES/PWN)
Time integrator	2nd-order TVD RK	forward to steady	3rd-order TVD RK	RK relaxation
Radiation	FLD , multi-band, implicit	XUV atten. + Ly α MC	Radial XUV atten. (53 bins)	Radial XUV atten.
Heating eff.	From C2Ray split	Photoelectron 20–40%	Freq.-avg. $\eta = 0.1–0.5$	Tuned efficiency
Ioniz. solve	C2Ray or implicit network	KPP preprocessor	Semi-implicit source	Newton / MINPACK
Metals	Via general chem.	Optional (solar)	None	C/N/O ioniz.+cooling
He(2 ³ S) 10830	Not a focus	In-loop non-LTE	Post-processed	In-loop + EXHALE_transit.py
H($n=2$)/H α	Not a focus	Non-LTE + Ly α MC	Post-processed	Non-LTE + Ly α esc. prob.

Axis	Aiolos	Taylor 25/26	Xing 23	Exhale
Cooling	Recomb/line/ff, dust, H_3^+	Recomb + H I line + H_3^+	Lyα only	Ly α +recomb+ff+CHIANTI
Tidal/Roche	USE_TIDES	Optional / off (2026)	Stellar tidal term	Roche (RLOF)
Lower boundary	Configurable	μbar, coupled	$1 R_p$, $T=1500$ K	$\sim \mu$ bar, fixed T, n
Transmission	—	Ray-traced Voigt	Post-processed	EXHALE.transit.py (multi-line)

4 What they have in common

1. **1-D spherical, XUV-driven escape** with photoionization heating balanced by Ly α /recombination cooling and adiabatic expansion is the shared physical backbone.
2. **Steady-state target.** All reach a stationary transonic wind — the three external codes by evolving the time-dependent equations forward, EXHALE by direct relaxation. The physical end state is the same.
3. **Tunable heating efficiency.** None derives the photoelectron heating efficiency from first principles; all treat it as the primary free knob, calibrated against He 10830 / mass-loss constraints.
4. **Same core H/He atomic physics.** Voronov collisional ionization, radiative + dielectronic recombination, and $H \leftrightarrow He$ charge exchange appear in Xing, Taylor, and EXHALE with near-identical rate coefficients (differing mainly in source table: Storey & Hummer vs Badnell vs Verner).
5. **He(2^3S) as the key observable.** Taylor, Xing, and EXHALE all target the 10830 Å line via a non-LTE 2^3S population (recombination, collisions, radiative decay, Penning ionization, photoionization). AIOLOS does not.
6. **Finite-volume, TVD-limited shock-capturing hydrodynamics** underlies AIOLOS, Xing (PLUTO), and EXHALE (WENO3). Taylor’s finite-difference thermosphere solver is the outlier.

5 Where they genuinely differ

5.1 Fluid model — the single most important axis

- **Xing is multi-fluid** (each species its own velocity), which reproduces **He/H mass fractionation**: because He is $4\times$ heavier, it is under-dragged by the H wind, so the *escaping* He/H (0.039) is $\sim$ half the bulk abundance (0.086) — explaining the weak 10830 without a subsolar He abundance.
- **Taylor is single bulk momentum but adds molecular + eddy diffusion**, which yields the *same* qualitative diffusive He/H separation ($8\% \rightarrow 2.5\%$ with altitude) at far lower cost; Taylor argues it reproduces Xing’s and Schulik & Owen (2025)’s multi-fluid results.
- **Aiolos is multi-fluid but escape-agnostic**: its friction solver would in principle yield fractionation, but the code is not tuned for the transonic-wind problem.
- **Exhale is strictly single-fluid** with a *fixed* He/H at all radii — it **cannot** represent fractionation or diffusive separation. This is the largest physical gap relative to the modern He 10830 literature. [2026-08-15: since implemented as the molecular + eddy diffusion term recommended below (`binary_element_diffusion.f90`), which transports the helium MASS FRACTION $X = \rho_{He}/\rho$ of a binary H/He mixture rather than the trace ratio n_{He}/n_H of

Taylor’s formulation; see the status note on page 1.]

5.2 Radiation transport

AIOLOS is the only one with a full **FLD** thermal-radiation solver (necessary for optically-thick interiors/dust, overkill for the escaping thermosphere). Taylor is the only one doing **Ly α Monte Carlo** RT, iterated with the hydro solution, for H($n=2$)/H α . Xing and EXHALE use simple radial XUV attenuation; EXHALE additionally treats Ly α trapping through a Neufeld escape-probability factor — a middle ground between Xing’s neglect and Taylor’s full Monte Carlo.

5.3 Ionization/chemistry solver

Taylor uses the KPP kinetic preprocessor (large stiff ODE network); AIOLOS uses C2Ray or a full implicit reaction matrix; Xing folds semi-implicit source terms into the PLUTO RK stages; EXHALE solves a single coupled *nonlinear algebraic* ionization system (Newton + analytic Jacobian, MINPACK fallback), appropriate for a steady-state code and already including trace metals in the coupled system.

5.4 Metals and cooling

EXHALE has the richest *closed-form* metal-cooling treatment (CHIANTI fits for Mg/Ca/Na/Fe II and C/N/O) and solves C/N/O ionization in the coupled system. Taylor can include solar-abundance metals but treats the structure as H/He-dominated (metals mainly lower the required heating efficiency). Xing has **no metals** and **only Ly α cooling**. AIOLOS carries cooling through general chemistry (Black 1981, H $_3^+$) but not specialized to the metal-line escape problem.

5.5 Line diagnostics

Taylor self-consistently computes both 10830 and H α in-loop with the most complete non-LTE microphysics (temperature-dependent Penning ionization, B-spline 2 3 S cross section, proton de-excitation, photoelectron impact excitation). Xing post-processes 10830 only. EXHALE builds the 2 3 S and H($n=2$) populations in-run and produces the full transmission spectrum (10830, Ly α , H α , H β , metal doublets) in `EXHALE_transit.py`. AIOLOS does not target these lines.

5.6 Lower boundary

Taylor couples to a photochemical lower/middle atmosphere at ~ 1 μ bar (the most physically motivated boundary). Xing and EXHALE use a fixed-density, fixed-temperature base near 1–2 μ bar. AIOLOS leaves the boundary fully configurable.

6 Adoption assessment for Exhale

EXHALE is already a dedicated, line-diagnostic-capable escape code, so it is closest to **Taylor** and **Xing** and furthest from **Aiolos** (whose value is a general RHD architecture EXHALE does not need). Each candidate element is rated by physics payoff and implementation cost, given EXHALE’s single-fluid steady-state architecture.

6.1 High payoff

(A) **He/H diffusive separation / mass fractionation — top priority.** This is EXHALE’s biggest physical gap and directly controls the He 10830 depth. Two routes:

- **Taylor route (recommended):** add a **molecular + eddy diffusion** term that lets He/H vary with radius while keeping the single bulk-momentum solve — a bolt-on to EXHALE’s continuity equations, moderate cost; Taylor shows it reproduces the multi-fluid answer, and it supplies the altitude-dependent He/H that `EXHALE.transit.py` needs.
- **Xing route (high cost):** convert EXHALE to genuinely multi-fluid (velocities for each species + collisional drag + electron-pressure Riemann split) — physically purest, but effectively a rewrite of the hydro core, hard to reconcile with EXHALE’s coupled-nonlinear ionization solve. **Not recommended** unless fractionation in the transonic/decoupled regime becomes a primary goal.

(B) **Temperature-dependent Penning ionization rate (Taylor 2025).** Taylor replaces the constant $\text{He}(2^3\text{S})+\text{H}$ Penning rate ($\sim 5 \times 10^{-10} \text{ cm}^3 \text{ s}^{-1}$) with a Maxwell–Boltzmann-averaged, temperature-dependent fit. Penning ionization is often the *dominant* 2^3S loss near the base, so this directly affects the 10830 prediction. **Low cost, high value** — a drop-in replacement; worth checking whether EXHALE currently uses the constant rate.

(C) **High-resolution $\text{He}(2^3\text{S})$ photoionization cross section (Taylor 2025).** Taylor’s B-spline K-matrix cross section resolves sharp EUV autoionization resonances that overlap stellar coronal lines. EXHALE has *already* moved this way (the recent TOPbase-based extension + two-VFKY96 + bridge in `cross_sec.f90`); Taylor’s machine-readable cross section is a natural cross-check / possible upgrade for the resonance region. **Low cost** (data swap), moderate value (resonance overlap is star-dependent).

6.2 Medium payoff

(D) **Whole-atmosphere lower-boundary coupling (Taylor).** Placing the base at $\sim 1 \mu\text{bar}$ with composition/temperature from a photochemical lower/middle atmosphere removes an ad hoc boundary. **Moderate-to-high cost**; valuable mainly for sub-Neptune / molecular regimes (GJ 1214b), less for the hot Jupiters EXHALE currently targets.

(E) **Molecular chemistry (H_2 , H_2^+ , H_3^+ , HeH^+) and H_3^+ cooling (Taylor 2026).** Needed only if EXHALE extends toward cooler/higher- μ sub-Neptunes, where H_2 survives into the wind and H_3^+ becomes a significant coolant. **Moderate cost**; defer unless the science scope broadens.

(F) **$\text{Ly}\alpha$ Monte Carlo vs Exhale’s Neufeld escape probability.** Taylor’s iterated Monte Carlo is the gold standard for the $\text{H}\alpha$ -driving $\text{Ly}\alpha$ field, but EXHALE already approximates trapping with a Neufeld escape probability (chosen because WASP-121b’s $\tau_0 \approx 5 \times 10^7$ makes brute-force Monte Carlo infeasible). Full Monte Carlo is **high cost for incremental gain**; keep the escape-probability approach, using Taylor’s setup as a validation benchmark.

6.3 Low payoff for Exhale

(G) **Aiolos’s FLD radiation, HLLC solver, general chemistry architecture.** Excellent for a *general* RHD code but solving problems EXHALE does not have: FLD matters for optically-thick

interiors/dust, not the optically-thin thermosphere; HLLC vs EXHALE’s approximate Riemann solver is not limiting for the smooth transonic wind; and EXHALE’s coupled-nonlinear ionization solve already suits steady state. **Not recommended** as targeted adoptions. AIOLOS is best used as an **independent cross-check** (its friction solver is an alternative reference for fractionation physics).

(H) Xing’s electron-pressure Riemann split. Elegant, but relevant only if EXHALE goes multi-fluid (route A/Xing), which is not recommended. **Skip** unless multi-fluid is pursued.

6.4 Recommended adoption order

1. **Temperature-dependent Penning ionization rate** (B) — cheapest, directly improves 10830.
2. **He/H diffusive separation via a diffusion term** (A, Taylor route) — largest physics gain for EXHALE’s core observable; keeps the single-fluid architecture.
3. **Cross-check / optionally upgrade the He(2^3S) cross section** against Taylor’s B-spline data (C) — EXHALE is already most of the way there.
4. Consider **whole-atmosphere boundary coupling** (D) and **molecular chemistry** (E) *only if* the scope extends to sub-Neptunes.
5. Treat **Aiolos and Xing multi-fluid** as **validation references**, not component donors.

7 Summary

- **Aiolos** is a general-purpose, multi-fluid RHD code with the most sophisticated *numerics* (HLLC, FLD, implicit chemistry) but the least specialization to the He/H α escape-diagnostic problem. Best used as an independent cross-check.
- **Taylor (2025/2026)** is the closest match to EXHALE’s goals and the richest source of *micro-physics* upgrades: temperature-dependent Penning ionization, a high-resolution He(2^3S) cross section, single-fluid diffusive He/H separation, non-LTE H($n=2$)/H α , and whole-atmosphere coupling.
- **Xing (2023)** provides the physically purest treatment of He/H mass fractionation via genuine multi-fluid dynamics, but at an architectural cost EXHALE should avoid; Taylor’s diffusion approximation captures most of the same effect far more cheaply.
- **Exhale’s single largest gap** is the absence of any radius-dependent He/H separation. Adopting a **Taylor-style diffusion term** (plus the **temperature-dependent Penning rate**) is the highest-value, lowest-risk path to bringing EXHALE’s He 10830 predictions in line with the current state of the art, without abandoning its single-fluid, steady-state, metal-aware design.